

UChicago Candidate Statement

Issey Sone

By earning a Master's of Statistics at the University of Chicago, I want to leverage rigorous statistical theory and modern computational tools to develop solutions to complex scientific problems. Currently, my interests lie in high-dimension and asymptotic statistics, with a particular focus on understanding the fundamental limits of inference as both sample size and dimensionality increase. I am especially drawn to modern, computationally scalable estimation procedures, and analyzing asymptotic properties such as convergence, bias, and uncertainty quantification under high-dimensional scaling. As an aspiring applied scientist, I am motivated by translating theoretical developments in probability and statistics into practical modeling tools for complex systems observed in real data.

My undergraduate studies at the University of Toronto have been some of the most rewarding and fulfilling experiences in my educational career. At UofT, I studied Statistics and Machine learning, with profound interests and research in probability theory. Throughout my undergraduate studies, I have developed a strong appreciation for statistical theory by engaging with proof-based coursework that emphasizes the mathematical foundations of statistics. I first began considering graduate studies in statistics when I finished my previous research work in the spring, and took a topic course in graduate-level probability in the summer. Through this course and the opportunity to present my work in a talk, I found that my strongest engagement came from the iterative process of research: working through technical derivations, confronting gaps in understanding, and refining ideas. These experiences motivated me to pursue graduate training in statistics. My main work has centered on theoretical probability, where I worked on problems involving limit theorems, large deviations, and Hypergeometric Functions like ${}_2F_1$.

Throughout this project, I have received early and rigorous feedback that challenged me to communicate my ideas with greater clarity and precision. Learning to respond constructively to this critique reshaped how I approach research: I became more deliberate in structuring arguments, and presenting results in a clear and compelling manner. As a result, I repeatedly revised my reports, developed clearer figures, transitioned from handwritten work to typesetting documents, and expanded documentation in code to improve clarity. While I came to realize the meaningful process in research stems from technical insights, these experiences shed light on the importance of persistence, revision, and a willingness to refine ideas in response to thoughtful critique. This research project has also taught me to be adaptable, and comfortable in new environments. The major part of the project involved heavy use of numerical analysis and work involving Mathematica. This is a specialized tool I had to utilize in order to ensure the best accuracy of approximation. While I had no prior experience in Mathematica, I had to be resourceful, often seeking documentation, setting up meetings with other colleagues, and engaging myself in the new environment. Learning to identify, understand, and integrate sources on my own significantly expanded my ability to initiate and drive research. It shifted my mindset to be more curious, less afraid of being unfamiliar, and trusting my profound problem-solving abilities.

In addition to theoretical work, I was involved in a joint applied project with OISE analyzing student behavior patterns using statistical modeling. Specifically, we wanted to investigate student behavior on an online learning platform. We asked questions like how long until a student will switch learning configuration? How can we model the transition for English usage on the platform for non-English speakers? How do transition rates differ for different speaker groups? To address these questions, I employed Markov chain models to estimate transition probabilities between language configurations while students engaged with online lectures. This modelling was suitable for its interpretability and clear-cut assumptions. While the approach provided a solid framework for understanding short-term transitions, the empirical complexity of student behavior, along with heterogeneity across users, limited the feasibility of fully capturing the dynamics with a simple Markov model. This experience helped me recognize both the strengths and limitations of interpretable statistical models, and reinforced the importance of carefully balancing model simplicity with inferential adequacy in applied settings. These limitations motivated my interest in flexible probabilistic approaches to high-dimensional inference, while also emphasizing the theoretical and computational challenges that emerge when dimensionality and complexity grow.

This project also shed light on the challenges of experimental design and gave me exposure to applied statistical modeling to answer empirical research questions. Collaborating with non-technical colleagues pushed me to communicate results through clear visualizations and to carefully justify modeling assumptions. These experiences strengthened my appreciation for rigorous, interpretable methods and further motivated my interest in applying computational and statistical techniques to real-world problems.

On top of research and projects, some of my most rewarding experiences come from teaching. The ability to teach is a rather complex one, one which requires skills like communication, depth of knowledge, and adaptability. My experience working as Teaching Assistant for both introductory Statistics, and high-level upper year Statistics courses have allowed me to refine my teaching for both basic and intricate ideas to a high standard. While explaining and breaking down difficult concepts is rewarding, it was also equally rewarding to be adaptable in explaining simpler concepts. I am deeply gratified when students express that my explanations are clear and that my communication supports their understanding, and can confidently attribute this to my strong depth of knowledge and my solid understanding of statistics and probability. I also voluntarily ran and organized 10+ review seminars for popular statistics courses offered at UofT through the departmental student association. Subsequently, as I commence my journey through graduate studies and beyond, I see teaching not as a parallel responsibility to research, but as an integral part of it. I hope to continue developing accessible instructional practices through transparent course design, flexible modes of engagement, and mentorship that acknowledges diverse paths into academia.

UChicago stands out as a compelling fit because of its strength in asymptotic theory and high-dimensional inference, reflected in its faculty expertise and interests. While I am particularly interested in areas of high-dimensional statistics, I am open to explore other various topics in statistical learning theory. Across the faculty's shared interests in high-dimensional statistics and inference, I believe I would be well positioned to develop the theoretical and computational tools needed to address problems of the modern world.

Ultimately, a Master's in Statistics at UChicago will train me to tackle intricate real-world problems as a researcher and applied scientist. UChicago's strength in high-dimensional statistics, supported by its faculty expertise and collaborative research culture, aligns closely with my goal of engaging in rigorous graduate-level research and deepening my formal training as a scientist. I am motivated by the opportunity to involve myself in UChicago's rigorous and highly engaged academic community, while contributing in return through teaching, seminars, and collaborative work. I am confident the mentorship I receive, combined with my dedication to my work, and preparation I developed through course rigor and previous research will allow me to excel at UChicago's Statistics department.